

Code-Layout, Readability and Reusability

Date	09-06-2026
Team ID	
Project Name	Automated Credit Card Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Wellorganized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Code indentation is consistent across all files	Yes	Consistent indentation maintained in app.py, Jupyter Notebook, and HTML templates.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized with app.py, credit_card_model.pkl, scaler.pkl, requirements.txt, templates/, static/, dataset/, and notebook files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as age, income, employment_status, credit_history, loan_amount, and existing_debt are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are used for data preprocessing, model loading, prediction, and result generation.
5	Code Comments	Inline and block comments explain logic	Partial	Key preprocessing, Machine Learning, and prediction functions are documented with comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules are maintained for preprocessing, model loading, prediction logic, and frontend templates.

7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable functions are implemented to minimize code duplication.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation and exception handling are implemented for prediction requests.



Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High/Medium/Low)
1	Credit Card Prediction Model	Python, scikitlearn	Credit card approval prediction	High
2	Data preprocessing & scaling module (StandardScaler)	Python, Scikit	Model training and prediction	High
3	Flask API	Python, Flask	Backend processing and prediction	High
4	HTML templates (Home / About / Find Your Crop / Result)	HTML, CSS	User Interface	Medium
5	Dataset loader	Python, Pandas	Training and testing process	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project is well organized with separate files for the ML model, dataset, application, and documentation.
Readability	5	Code uses meaningful variable names, proper indentation, and clear function names.
Reusability	5	Reusable modules like the crop prediction model, dataset loader, and preprocessing/scaling are used across the project.
Documentation / Comments	4	Basic comments and project documentation are provided, but more detailed inline documentation can be added.
Overall Score	5	The project follows good coding standards, modular design, and reusable components with a clean structure.